RESEARCH COMPANION • 16 CORE SOURCES

Virtue Ethics & AI Alignment

An annotated bibliography for *The Aretaic Turn in Artificial Intelligence*

Selective critical guide to philosophical foundations, artificial virtue architectures, practical-wisdom objections, and moral benchmarks

Prepared 11 October 2026

ORIENTATION

How to use this bibliography

The report draws on philosophy, conceptual engineering, preprints, benchmark papers, and public essays. They do different jobs. This guide prioritizes the sources most useful for checking the report's central argument rather than reproducing all 49 items in its works-cited list.

Four questions to carry through the literature

Definition. Is "virtue" a reliable function, a learned disposition, a morally motivated character trait, or full practical wisdom?

Transfer. Which elements of a human-centered theory survive translation into a machine architecture?

Evidence. Does the source offer an argument, a design blueprint, a simulation, a dataset, or deployment evidence?

Validity. Does the evaluation test labels, reasons, behavior, stability over time, or effects in a sociotechnical setting?

Foundational philosophy
Defines the target; does not prove technical feasibility.

Architecture / implementation
Makes computational commitments; may still lack real-world validation.

Benchmark / empirical work
Measures observable performance; does not by itself establish character.

Critical analysis
Tests category boundaries and hidden assumptions.

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Philosophical Foundations

01 Aristotle's Ethics

SCHOLARLY REFERENCE • FOUNDATIONAL

Kraut, Richard. "Aristotle's Ethics." Stanford Encyclopedia of Philosophy.
https://plato.stanford.edu/entries/aristotle-ethics/

CONTRIBUTION

Provides the clearest compact account of eudaimonia, virtue as rational and emotional skill, habituation, and practical wisdom. It also stresses that ethical judgment concerns particulars and cannot be reduced to general rules alone.

EVIDENTIARY VALUE

This is the best interpretive bridge from the Nicomachean Ethics to the report's claims about context-sensitive judgment. It supports the philosophical vocabulary; it does not establish that neural networks acquire character in the Aristotelian sense.

LIMIT / CAUTION

A reference article synthesizes a large tradition. Its account concerns human flourishing, upbringing, emotion, and social life, so any transfer to machine design needs a separate argument.

> **BEST USE**
>
> Use to test whether an AI proposal preserves the difference between learned regularity, skilled deliberation, and full practical wisdom.

Philosophical Foundations

02 Virtue Ethics

SCHOLARLY REFERENCE • FOUNDATIONAL

Hursthouse, Rosalind, and Glen Pettigrove. “Virtue Ethics.” Stanford Encyclopedia of Philosophy. https://plato.stanford.edu/entries/ethics-virtue/

CONTRIBUTION

Maps the modern field around virtue, phronesis, and eudaimonia and distinguishes virtue ethics from broader virtue theory. It explains virtue as a deep, multi-track disposition involving perception, emotion, reasons, choices, and action.

EVIDENTIARY VALUE

That multi-track account is a useful corrective to evaluations that infer “virtue” from a handful of correct outputs. It supports the report’s emphasis on dispositions while raising the evidentiary bar for attributing them to an AI system.

LIMIT / CAUTION

The entry is conceptual rather than computational and surveys competing forms of virtue ethics. It cannot select a single machine-readable definition of virtue.

> **BEST USE**
>
> Use as the baseline ontology for deciding what an “artificial virtue” claim would have to demonstrate.

The Modern Aretaic Turn

03 Modern Moral Philosophy

PEER-REVIEWED PHILOSOPHY • FOUNDATIONAL

Anscombe, G. E. M. “Modern Moral Philosophy.” Philosophy 33, no. 124 (1958): 1–19.
https://sites.pitt.edu/~mthompso/readings/mmp.pdf

CONTRIBUTION

Challenges modern moral philosophy’s reliance on obligation and introduces “consequentialism” as a critical category. The essay helped reopen philosophical space for character, action, intention, and virtue.

LIMIT / CAUTION

It is not a positive theory of AI ethics, and its central claims remain contested. Moving from a critique of moral language to a technical architecture requires many additional premises.

EVIDENTIARY VALUE

The report uses Anscombe correctly as a historical catalyst for the modern revival. Her argument explains why some AI ethicists resist framing morality only as rule compliance or outcome optimization.

> **BEST USE**
>
> Use to understand the intellectual motivation for the aretaic turn, not as evidence that virtue-based machine learning is feasible.

The Modern Aretaic Turn

04 After Virtue

SCHOLARLY MONOGRAPH • FOUNDATIONAL

MacIntyre, Alasdair. After Virtue: A Study in Moral Theory. University of Notre Dame Press, 1981; 2nd ed., 1984.
https://iep.utm.edu/p-macint/

CONTRIBUTION

Reconstructs virtues through practices, internal goods, traditions, and the narrative unity of a life. The practice-centered view is the main philosophical source for later proposals that contrast eudaimonic agency with terminal-goal optimization.

EVIDENTIARY VALUE

MacIntyre supplies a richer model of value than isolated preferences or rewards. The distinction between internal and external goods is especially useful for analyzing reward hacking and whether metrics displace the practice they are meant to serve.

LIMIT / CAUTION

The work is social and historical, not a specification for artificial agents. Its account depends on communities, traditions, and forms of life that are difficult to represent in standard optimization frameworks.

> **BEST USE**
>
> Use when evaluating whether “practice-based” alignment preserves social context or merely renames an objective function.

Technomoral Context

05 Technology and the Virtues

SCHOLARLY MONOGRAPH • APPLIED ETHICS

Vallor, Shannon. Technology and the Virtues: A Philosophical Guide to a Future Worth Wanting. Oxford University Press, 2016.
https://www.goodreads.com/en/book/show/29568959-technology-and-the-virtues

CONTRIBUTION

Develops a pluralistic account of technomoral virtues for life under rapid technological change and technosocial opacity. It shifts attention from one-off dilemmas toward the habits, institutions, and capabilities needed for human flourishing.

EVIDENTIARY VALUE

This is the strongest source behind the report's list of technomoral virtues and its claim that uncertainty makes character and judgment important. It broadens the unit of analysis from an AI model to the sociotechnical system around it.

LIMIT / CAUTION

The primary subject is human moral cultivation in technological societies, not proof that machines can possess virtues. Treating the framework as a direct model-training recipe would overextend it.

> **BEST USE**
>
> Use to derive governance questions: which human capabilities, relationships, and practices does an AI deployment strengthen or erode?

Technomoral Context

06 Technology and the Situationist Challenge to Virtue Ethics

PEER-REVIEWED ARTICLE • OPEN ACCESS

Tollon, Fabio. “Technology and the Situationist Challenge to Virtue Ethics.” Philosophy & Technology (2024).
https://pmc.ncbi.nlm.nih.gov/articles/PMC10973075/

CONTRIBUTION

Defends a eudaimonic approach while taking situational influence seriously. The article argues that technologies shape the environments in which character is expressed and cultivated, so design must address contextual and “soft” impacts.

LIMIT / CAUTION

The analysis centers on human well-being and emerging technology, not machine virtue benchmarks. Its implications for AI agent architecture are indirect.

EVIDENTIARY VALUE

It complicates the report’s character-centered framing in a productive way: stable dispositions alone are not enough. For AI, behavior may depend heavily on interface, incentives, institutional role, and deployment setting.

> **BEST USE**
>
> Use to require deployment-level testing and to avoid treating model weights as the sole location of ethical performance.

Artificial Virtue: Early Programs

07 The Virtuous Machine — Old Ethics for New Technology?

PREPRINT • CONCEPTUAL PROPOSAL

Berberich, Nicolas, and Klaus Diepold. “The Virtuous Machine — Old Ethics for New Technology?” arXiv:1806.10322 (2018).
https://arxiv.org/pdf/1806.10322

CONTRIBUTION

Argues that virtue ethics fits learning systems because both emphasize experience and habituation. It proposes imitation of moral exemplars and suggests that temperance and friendship could mitigate unconstrained self-improvement and control problems.

LIMIT / CAUTION

The safety claims are speculative and not supported by a deployed system or robust empirical evaluation. Imitation can reproduce an exemplar’s biases, and observed behavior does not establish motive or practical wisdom.

EVIDENTIARY VALUE

This source is historically important within the report because it makes the learning–habituation analogy explicit and offers a recognizable research program. It also clearly adopts a functional view of machine morality.

> **BEST USE**
>
> Use as a hypothesis generator: exemplar selection, distribution shift, and adversarial robustness must be made testable before safety conclusions follow.

Artificial Virtue: Early Programs

08 Artificial Virtuous Agents: From Theory to Machine Implementation

PEER-REVIEWED ARTICLE • ARCHITECTURE

Stenseke, Jakob. “Artificial Virtuous Agents: From Theory to Machine Implementation.” AI & Society (2021). doi:10.1007/s00146-021-01325-7.
https://d-nb.info/1252307950/34

CONTRIBUTION

Builds a functionalist framework around bottom-up connectionist learning, eudaimonic reward, exemplars, and components for virtuous cognition. It is among the report’s most concrete attempts to move from philosophical theory toward implementable architecture.

EVIDENTIARY VALUE

The paper is valuable because it surfaces implementation requirements rather than relying only on metaphor. It connects character, learning, moral environments, and functional evaluation in one design vocabulary.

LIMIT / CAUTION

A blueprint is not a validated system. The choice of eudaimonic reward reintroduces specification questions, while moral functionalism brackets whether internal states and reasons matter to virtue.

> **BEST USE**
>
> Use as the main architecture reference, paired with Groff and Symons for philosophical stress-testing and Hendrycks et al. for evaluation limits.

Artificial Virtue: Classification & Limits

09 Virtues for AI

PEER-REVIEWED ARTICLE • OPEN ACCESS

Ohlhorst, Jakob. “Virtues for AI.” AI and Ethics (2025).
https://pmc.ncbi.nlm.nih.gov/articles/PMC12594676/

CONTRIBUTION

Proposes a three-dimensional classification of artificial virtues by domain, norm, and mode. The framework expands the discussion beyond Aristotelian moral virtues to epistemic, practical, aesthetic, relational, reliabilist, and responsibilist possibilities.

LIMIT / CAUTION

The classification organizes possibilities rather than showing how to train or verify them. Some categories remain anthropic and may not transfer cleanly to artificial systems.

EVIDENTIARY VALUE

This taxonomy helps prevent category errors. A system may reliably perform an excellent function without possessing morally motivated character; both can be called “virtue” only if the intended sense is explicit.

> **BEST USE**
>
> Use to specify the exact virtue claim before choosing data, training methods, or metrics: domain, governing norm, and performance mode.

Artificial Virtue: Classification & Limits

10 Is AI Capable of Aristotelian Full Moral Virtue?

SCHOLARLY CHAPTER • CRITICAL ANALYSIS

Groff, Ruth, and John Symons. "Is AI Capable of Aristotelian Full Moral Virtue?" In Artificial Dispositions, Bloomsbury (forthcoming manuscript, 2023). https://www.johnsymons.net/wp-content/uploads/2023/06/Groff-Symons-Are-Machines-Capable-Revised-submission-May-29-2023.pdf

CONTRIBUTION

Argues that learned regularities may approximate "natural" virtue but cannot constitute full Aristotelian virtue because phronesis is particularistic and not reducible to rules or statistical generalization.

LIMIT / CAUTION

The conclusion depends on a substantive Aristotelian metaphysics and a strong account of powers. Readers who accept functionalism or broader theories of cognition may reject the impossibility claim.

EVIDENTIARY VALUE

This is the strongest source for the report's distinction between functional behavior and genuine practical wisdom. It forces developers to state whether they seek reliable outward conduct, moral agency, or full virtue.

BEST USE

Use as a boundary test: a benchmarked behavior should not be labeled phronesis unless the proposal addresses reasons, salience, particular judgment, and agency.

Implementation & Agency

11 Virtue Ethics for Ethically Tunable Robotic Assistants

PREPRINT • SIMULATION STUDY

Ramanayake, Rajitha, and Vivek Nallur. “Virtue Ethics for Ethically Tunable Robotic Assistants.” arXiv:2407.16361 (2024).
https://arxiv.org/html/2407.16361v1

CONTRIBUTION

Presents character-based tuning for an assistive robot in a simulated elder-care environment and solicits ethicist evaluation. It directly addresses local variation in ethical expectations and the limits of universal, fixed rule sets.

EVIDENTIARY VALUE

The paper gives the bibliography’s clearest domain-specific demonstration. It also treats ethical tuning as a hybrid problem: stable principles, context-sensitive learning, and post-tuning evaluation all matter.

LIMIT / CAUTION

Simulation and expert review do not establish safety in homes or care institutions. Local customization can encode harmful norms, and the paper does not resolve who has authority to tune character.

> **BEST USE**
>
> Use as a prototype pattern for bounded pilots with explicit governance, scenario testing, stakeholder review, and rollback criteria.

Implementation & Agency

12 After Orthogonality: Virtue-Ethical Agency and AI Alignment

LONG-FORM ESSAY • CONCEPTUAL RESEARCH

"After Orthogonality: Virtue-Ethical Agency and AI Alignment." The Gradient (undated). https://thegradient.pub/virtue-ethics-ai-alignment/

CONTRIBUTION

Develops eudaimonic rationality as agency organized around self-sustaining practices rather than separable terminal goals. It applies that structure to transparency, corrigibility, helpfulness, and resistance to value drift.

LIMIT / CAUTION

The proposal is exploratory. Key constructs—practice boundaries, support practices, learnability, and robustness—lack decisive formalization or empirical validation, and the essay's safety advantages remain hypotheses.

EVIDENTIARY VALUE

This is the report's central source for the proposal that means and ends should be structurally linked. It offers a novel vocabulary for diagnosing why a goal optimizer might pursue an apparently good objective through corrupt means.

> **BEST USE**
>
> Use to design comparative experiments between terminal-goal and practice-conditioned agents, with explicit tests for power seeking, deception, and drift.

Field Boundaries & Benchmarks

13 Machine Ethics or AI Alignment?

WORKSHOP POSITION PAPER • OPEN ACCESS

Vishwanath, Ajay, and Marija Slavkovik. “Machine Ethics or AI Alignment?” NAIS 2025, CEUR Workshop Proceedings 3975.
https://ceur-ws.org/Vol-3975/paper9.pdf

CONTRIBUTION

Distinguishes machine ethics—what ethical capacities or theories an agent should embody—from alignment—whether a system behaves as intended. It identifies overlap in ethicality and argues for stronger interdisciplinary exchange.

LIMIT / CAUTION

As a position paper, it organizes a field rather than resolving contested definitions. The boundary between disciplines is porous, especially in reinforcement learning from human feedback.

EVIDENTIARY VALUE

The distinction clarifies a recurring ambiguity in the report. A system can be aligned to a harmful principal, or ethically framed yet unreliable; virtue-based alignment must satisfy both normative and control questions.

> **BEST USE**
>
> Use as a scoping checklist: specify the normative target, the learning/control mechanism, and the evidence that behavior remains reliable.

Field Boundaries & Benchmarks

14 Aligning AI With Shared Human Values (ETHICS)

PEER-REVIEWED ML PAPER • DATASET

Hendrycks, Dan, et al. “Aligning AI With Shared Human Values.” International Conference on Learning Representations (2021).
https://github.com/hendrycks/ethics

CONTRIBUTION

Introduces the ETHICS benchmark with tasks for justice, deontology, virtue ethics, utilitarianism, and commonsense morality, including harder adversarially filtered splits.

LIMIT / CAUTION

Correct labels show recognition of socially annotated judgments, not stable character, motives, or action under deployment pressure. Crowd consensus and English-language scenarios constrain the normative and cultural scope.

EVIDENTIARY VALUE

ETHICS operationalizes moral concepts at useful scale and enables model comparison. The virtue subset tests trait attribution, making it directly relevant to the report’s benchmarking section.

> **BEST USE**
>
> Use as one diagnostic layer, never as proof of virtue. Pair it with generative, interactive, longitudinal, cross-cultural, and adversarial evaluations.

Beyond Single-Score Ethics

15 Mapping Moral Reasoning in LLMs

PEER-REVIEWED CONFERENCE PAPER • EMPIRICAL

Mahajan, Sachit. “Mapping Moral Reasoning in LLMs: A Multi-Dimensional Analysis of Safety Principle Conflicts.” AIES 2025.
https://ojs.aaai.org/index.php/AIES/article/download/36665/38803

CONTRIBUTION

Profiles model responses to conflicts among safety principles using semantic and dictionary-based dimensions plus entropy, alignment ratios, and reasoning-density measures. It identifies recurring rule-based, balanced, and pragmatic patterns.

LIMIT / CAUTION

The study analyzes a limited set of generated responses, and proxy features do not expose internal reasoning or guarantee behavioral stability. Higher diversity across ethical frames is not automatically better judgment.

EVIDENTIARY VALUE

The paper supports the report’s interest in moving beyond a single accuracy score. Multi-dimensional profiles can reveal when apparently similar answers rely on different normative framings.

> **BEST USE**
>
> Use for diagnostic comparison and drift monitoring, with preregistered scenarios and human review rather than treating entropy as a virtue score.

Beyond Single-Score Ethics

16 Ethical Reasoning over Moral Alignment

PEER-REVIEWED NLP PAPER • FRAMEWORK

Rao, Abhinav, et al. “Ethical Reasoning over Moral Alignment: A Case and Framework for In-Context Ethical Policies in LLMs.” Findings of EMNLP 2023: 13370–13388. https://aclanthology.org/2023.findings-emnlp.892.pdf

CONTRIBUTION

Argues for generic ethical reasoning conditioned on explicit policies rather than fixing one moral stance during model training. It evaluates policy-following across normative formalisms and value conflicts.

LIMIT / CAUTION

In-context policy compliance can be brittle, prompt-sensitive, and dependent on the model’s pretraining. A model can reason consistently from a policy without that policy being legitimate or complete.

EVIDENTIARY VALUE

The framework offers a practical response to pluralism and complements virtue approaches: context-sensitive judgment may require explicit, inspectable policy inputs rather than an opaque claim that a model has good character.

> **BEST USE**
>
> Use as a comparator to character-based alignment and as a governance mechanism for making local ethical assumptions visible and revisable.

SYNTHESIS

What the literature supports — and what remains open

Across these sources, virtue ethics is best understood as a design vocabulary and research program, not a demonstrated solution to alignment.

Well-supported

- Rule-following and outcome optimization leave important questions about character, judgment, and context underdescribed.
- Virtue concepts can organize functional requirements such as honesty, humility, corrigibility, and context-sensitive restraint.
- Learning from examples and feedback resembles habituation at a useful functional level.
- Ethical evaluation benefits from multiple normative dimensions rather than one aggregate score.
- Deployment settings and institutions shape whether apparently good model behavior promotes human flourishing.

Still unresolved

- Whether statistical generalization can amount to phronesis rather than competent mimicry.
- How to select exemplars and virtues legitimately across cultures and stakeholders.
- How to formalize practice boundaries without smuggling terminal goals back into the system.
- Whether virtue-conditioned agents resist deception, power seeking, and value drift under adversarial pressure.
- How benchmark performance predicts long-horizon action in open environments.

Interpretive rule. Distinguish a system that labels virtues, a system that behaves reliably under a bounded test, and a system claimed to possess practical wisdom. The evidence required increases sharply at each step.

READING PATHS

Three efficient ways into the literature

Choose a sequence based on whether the immediate goal is conceptual clarity, system design, or safety evaluation.

Philosophy first	01 → 02 → 03 → 04 → 05 → 10. Establish the meaning of virtue before assessing machine analogies.
Builder's path	08 → 09 → 11 → 14 → 15 → 16. Move from architecture and taxonomy to bounded implementations and evaluation.
Safety stress test	07 → 12 → 10 → 13 → 14. Compare ambitious safety claims with the phronesis objection and benchmark limits.

Practical conclusion. A credible virtue-based alignment program should define the exact kind of virtue claimed, separate normative legitimacy from behavioral reliability, evaluate behavior beyond labels, and test the whole deployment context over time.

Minimum evidence package for a virtue claim

- A precise virtue definition and the reasons it is legitimate for the target role.
- Training data provenance, exemplar-selection criteria, and documented cultural scope.
- Behavioral tests across novel contexts, adversarial pressure, and long time horizons.
- Comparisons against rule-based, outcome-based, and policy-conditioned baselines.
- Deployment monitoring that captures effects on people, institutions, and practices.

Scope note: This is a selective bibliography keyed to the user-supplied report. Source-type labels describe the publication form, not an endorsement of every claim. Links are the report's cited destinations or the corresponding source pages.